Operating Systems – Unit 4

Orange Program Assignment

Name: Sharana Basava

SRN: PES2UG24CS457

Section: H

Course: Operating Systems

Faculty: Dr. Gokulakrishnan S

Repo screenshot:

Code Issues Pull requests Agents Actions Projects

PES2UG24CS457-pes-vcs Public

main 1 Branch 0 Tags

Sharanabasava12 Initial commit

.gitignore	Initial cor
Makefile	Initial cor
README.md	Initial cor
commit.c	Initial cor
commit.h	Initial cor
index.c	Initial cor
index.h	Initial cor
object.c	Initial cor
pes.c	Initial cor
pes.h	Initial cor
test_objects.c	Initial cor
test_sequence.sh	Initial cor

Build success:

```
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$ make clean
make all
rm -f pes test_objects test_tree object.o tree.o index.o com
rm -rf .pes
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$
```

Test_objects:

```
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea092
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$
```

Test_tree:

```
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$
```

Test_sequence:

```
All Phase 2 tests passed!
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$ bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Commit created successfully

Log after first commit:
commit 337f4e3ad9172645b12a0c21aaa596b13316b56e33ccc70b7fe876ec90eb92b7

--- Second Commit ---
Commit created successfully

--- Third Commit ---
Commit created successfully

--- Full History ---
commit ced495654b97629383f7f7b302848d0934f9fec8e7bfe2b855d618a6117cae91

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
ced495654b97629383f7f7b302848d0934f9fec8e7bfe2b855d618a6117cae91

--- Object Store ---
Objects created:
8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/1f/2bee2143c6d0cb30beff8d73f6be8aa6b6d9590fe3418d1c25e455e1c17122
.pes/objects/33/7f4e3ad9172645b12a0c21aaa596b13316b56e33ccc70b7fe876ec90eb92b7
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/ce/d495654b97629383f7f7b302848d0934f9fec8e7bfe2b855d618a6117cae91
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$
```

Git proof:

```
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$ git commit -m "Completed orange assignment"
git push origin main
git log --oneline
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: empty ident name (for <sharanabasava@victus.localdomain>) not allowed
fatal: unable to access 'https://github.com/Sharanabasava12/PES2UG24CS457-pes-vcs.git/': Could not resolve host: github.com
74e1d20 (HEAD -> main, origin/main, origin/HEAD) Initial commit
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$
```

Pes init:

```
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$ ./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Commit created successfully
Commit created successfully
Commit created successfully
commit ced495654b97629383f7f7b302848d0934f9fec8e7bfe2b855d618a6117cae91
```

Make pes:

```
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$ make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
cat .pes/index    # Human-readable text format
make: 'pes' is up to date.
Initialized empty PES repository in .pes/
Staged changes:
  staged:     hello.txt
  staged:     bye.txt
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  tree.c
  untracked:  commit.h
  untracked:  pes.h
  untracked:  pes.c
  untracked:  index.c
  untracked:  commit.c
  untracked:  tree.h
  untracked:  test_objects.c
  untracked:  object.c
  untracked:  test_sequence.sh
  untracked:  README.md
  untracked:  index.h
  untracked:  test_tree
  untracked:  test_objects
  untracked:  Makefile
  untracked:  .gitignore
  untracked:  test_tree.c

100644 05c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43 1776273814 12 hello.txt
100644 e67ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc 1776273814 8 bye.txt
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776273841 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776273841 6 file2.txt
sharanabasava@victus:~/PES2UG24CS457-pes-vcs$ make pes
```